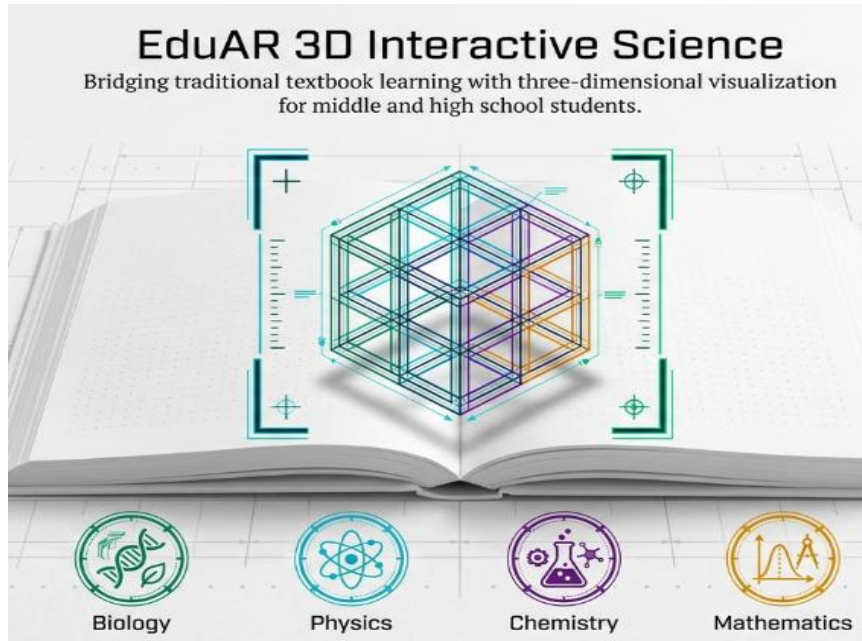


EDUCATIONAL AR SCIENCE BOOK

EduAR 3D

Interactive Science Through Augmented Reality



SCAN TO DOWNLOAD THE EDUAR 3D APP

(sample / placeholder code — replace with your live app-store link)

Target learners: Middle and high school students

BIOLOGY	PHYSICS	CHEMISTRY	MATHEMATICS
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INTRODUCTION

About This Book

EduAR 3D is an interactive science learning book designed to work together with a mobile augmented reality application.

The purpose of this book is to connect traditional textbook learning with interactive 3D visualization.

Students do not only read about a scientific concept. They can also use a smartphone to scan selected images in the book and explore a corresponding three-dimensional model.

The Learning Process

READ	SCAN	EXPLORE	UNDERSTAND	TEST
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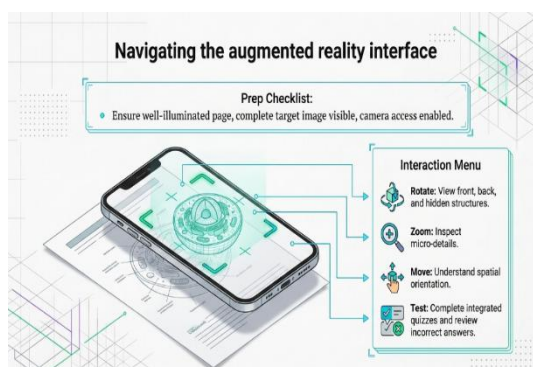
How the AR Target Works

Each AR-enabled lesson contains a specially designed image called an EduAR 3D Target.

When the student points the EduAR 3D mobile application at the target image, the corresponding 3D model appears in the camera view.

Students can interact with the model by:

- Rotating it
- Zooming in and out
- Moving the model
- Inspecting important structures
- Reading explanations
- Completing a short quiz



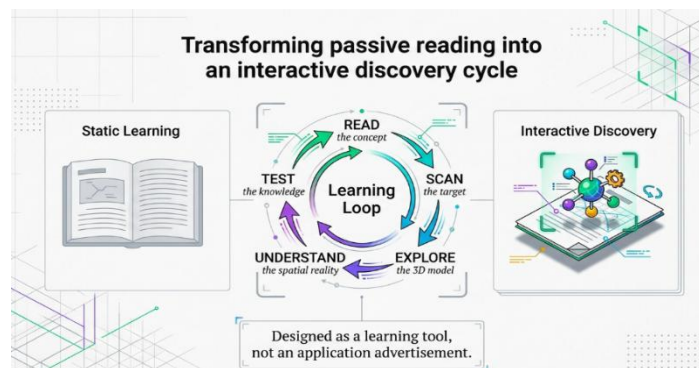
i A Note to Teachers and Students

The book should be used as a learning tool, not as an advertisement for the mobile application. Every lesson combines printed scientific explanation with a hands-on AR exploration activity.

PART 1

Biology

Human Anatomy



Biology is the study of living organisms and the systems that allow them to survive and function.

In this section, students will explore three important human organs, each with a dedicated AR target image:

- Human Heart
- Human Brain
- Human Lungs

LESSON 1

The Human Heart

Introduction

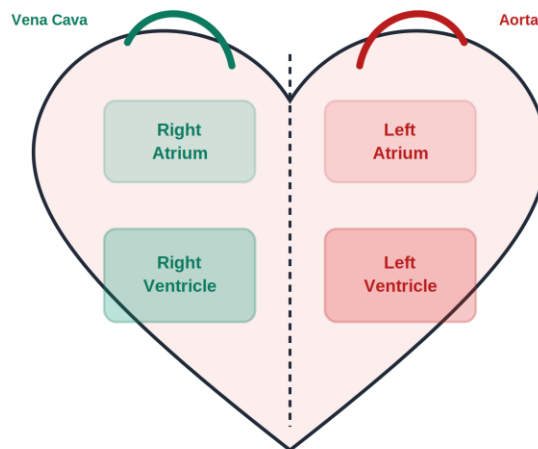
The human heart is a muscular organ that works continuously to circulate blood throughout the body.

Blood carries oxygen and nutrients to tissues and also helps transport waste products away from cells.

The heart is located in the chest between the lungs and is protected partly by the rib cage.

The human heart has four main chambers that work together to maintain continuous blood circulation:

- Right atrium
- Right ventricle
- Left atrium
- Left ventricle



Simplified diagram — four chambers of the human heart

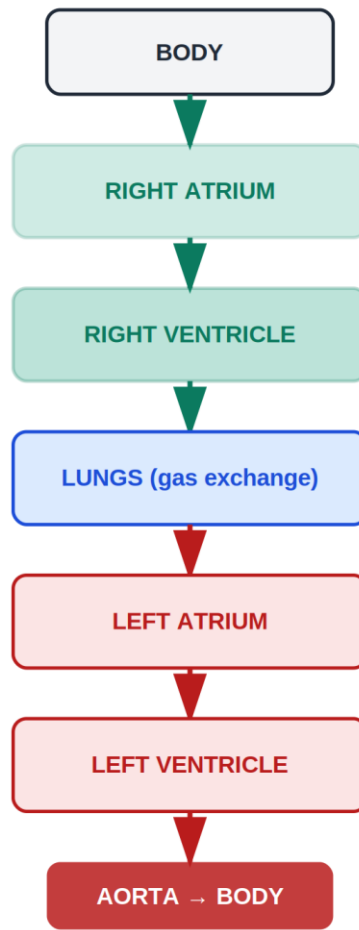
Diagram — the four chambers of the human heart

How Blood Moves Through the Heart

Deoxygenated blood returning from the body enters the right atrium, then moves into the right ventricle. The right ventricle pumps the blood toward the lungs through the pulmonary arteries.

In the lungs, the blood releases carbon dioxide and receives oxygen. The oxygen-rich blood then returns to the heart through the pulmonary veins, entering the left atrium and moving into the left ventricle.

The left ventricle pumps the oxygen-rich blood into the aorta, which distributes it throughout the body.



Simplified circulation pathway

Important Heart Structures

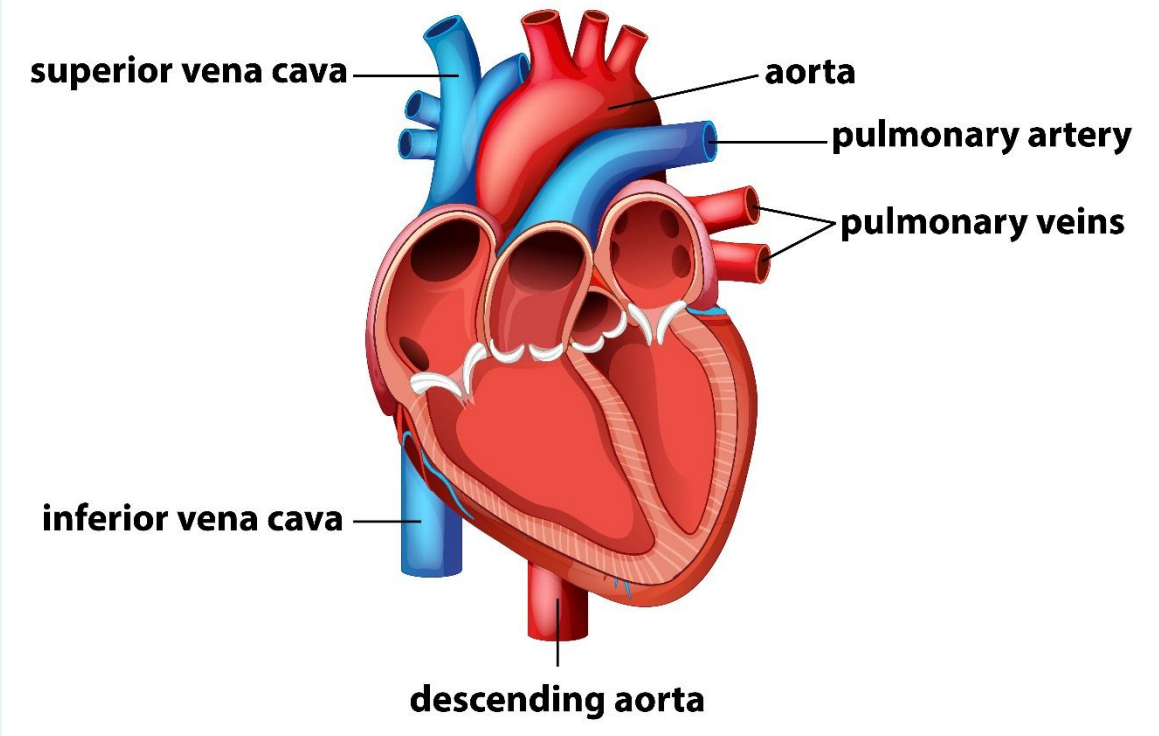
Right Atrium	Receives oxygen-poor blood returning from the body.
Right Ventricle	Pumps oxygen-poor blood toward the lungs.
Left Atrium	Receives oxygen-rich blood from the lungs.
Left Ventricle	Pumps oxygen-rich blood to the rest of the body.
Aorta	The major artery carrying oxygen-rich blood from the left ventricle to the body.
Vena Cava	Returns oxygen-poor blood from the body to the right side of the heart.
Heart Valves	Help maintain one-way blood flow through the heart.

EduAR 3D Activity

EDUAR 3D TARGET

Insert the Human Lungs AR target image here

Human Heart Blood Vessels



EduAR 3D Target — Scan this image to explore the human heart

Open the EduAR 3D mobile application and point the camera at the target image. When the target is recognized, the three-dimensional heart model will appear.

Explore

1. Rotate the heart to view the front and back.
2. Zoom into the chambers.
3. Locate the right atrium.
4. Locate the left ventricle.
5. Identify the aorta.
6. Follow the direction of blood flow.

Think About It

- Why does the left ventricle need a strong muscular wall?
- How is oxygen-poor blood different from oxygen-rich blood?
- Why are valves important?

Quick Review

7. How many chambers does the human heart have?
8. Which chamber pumps blood to the lungs?
9. Which chamber pumps oxygen-rich blood to the body?
10. Which major artery carries blood away from the left ventricle?

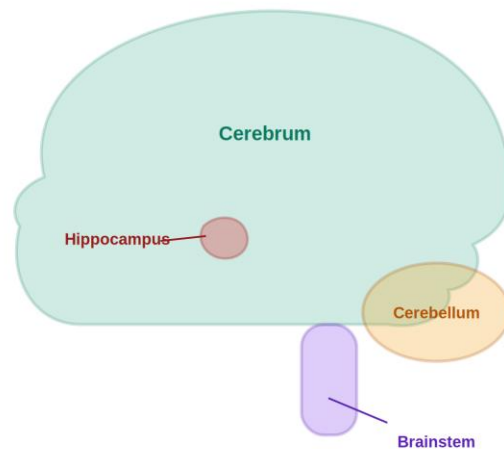
LESSON 2

The Human Brain

Introduction

The brain is the main control center of the nervous system. It receives information from the senses, processes information, controls movement, contributes to learning and memory, and helps coordinate many functions of the body.

The brain is protected by the skull. Although the brain appears as one organ, it contains many specialized regions that work together.



Simplified diagram — major regions of the human brain

Diagram — major regions of the human brain

Major Brain Structures

Cerebrum

The cerebrum is the largest part of the human brain. It is involved in:

- Thinking
- Reasoning
- Learning
- Memory
- Language
- Voluntary movement
- Processing sensory information

Cerebellum

The cerebellum is located near the back and lower part of the brain. It contributes to:

- Balance
- Coordination
- Precision of movement

Brainstem

The brainstem connects the brain with the spinal cord. It is involved in important automatic functions such as breathing and heart-rate regulation.

Hippocampus

The hippocampus is a brain structure strongly associated with learning and memory.

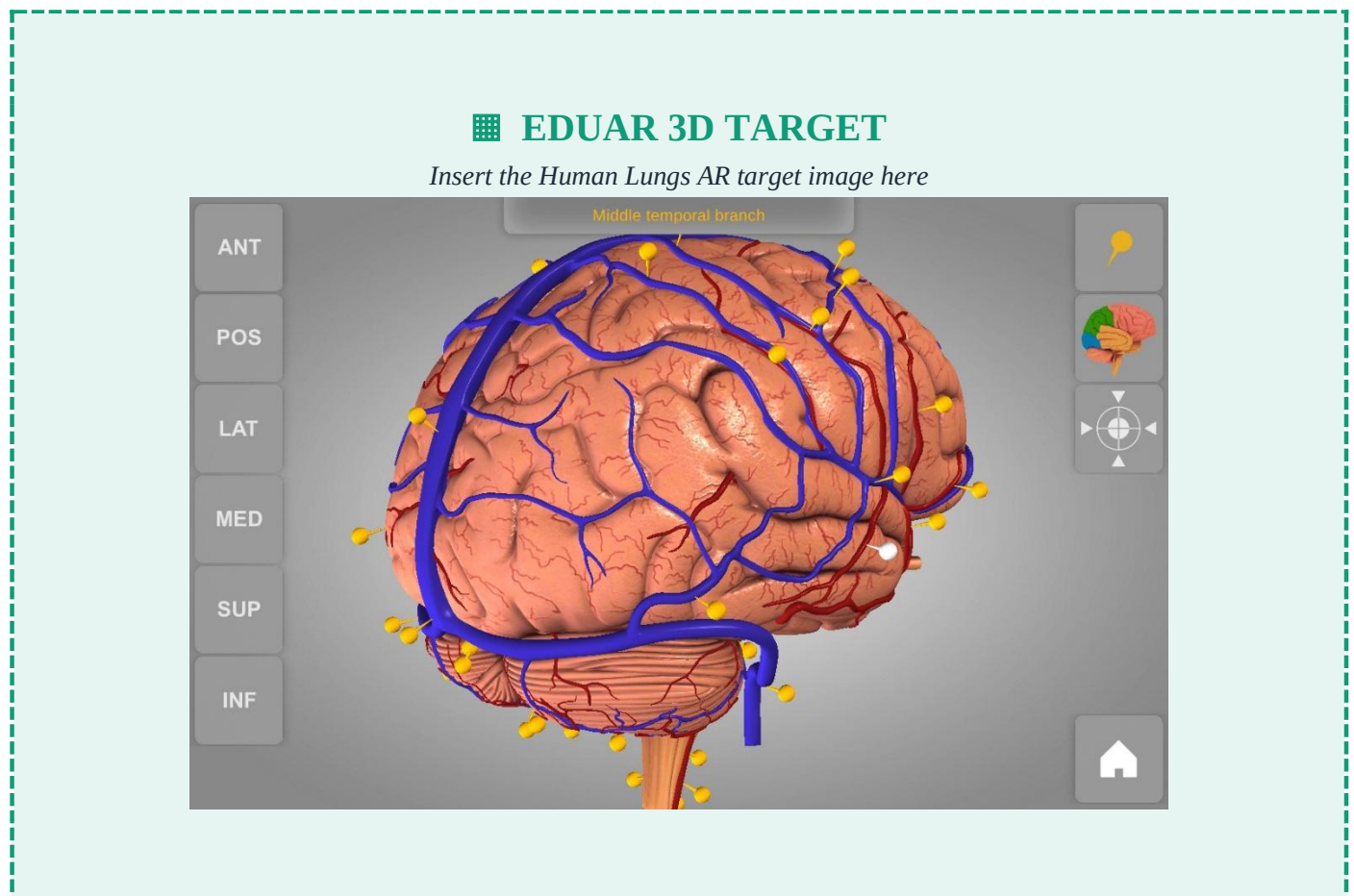
Why the Brain Is Important

The human body constantly receives information from the environment. For example, the eyes detect light and color, the ears detect sound, and the skin detects touch and temperature.

This information is processed by the nervous system. The brain interprets information and coordinates responses, allowing people to:

- Learn
- Remember
- Communicate
- Move
- Make decisions
- Respond to the environment

EduAR 3D Activity



EduAR 3D Target — Scan this image to explore the human brain

Explore

11. Rotate the brain.
12. View its different regions.
13. Locate the cerebrum.
14. Identify the cerebellum.
15. Find the brainstem.
16. Observe the overall shape of the brain.

Challenge

Can you identify the difference between the cerebrum and cerebellum?

Think About It

- Why is the brain protected by the skull?
- Why are different regions of the brain specialized for different functions?
- How could damage to the cerebellum affect movement?

LESSON 3

The Human Lungs

Introduction

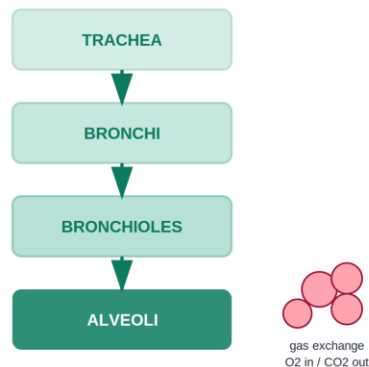
The lungs are the main organs of the respiratory system. Their major function is gas exchange. During breathing, oxygen enters the body and carbon dioxide is removed.

The respiratory system works together with the circulatory system to provide oxygen to cells throughout the body.

Airflow Through the Respiratory System

Air enters through the nose or mouth and travels through the trachea, bronchi, bronchioles, and finally the alveoli — tiny air sacs where gas exchange takes place.

Oxygen moves from the air inside the alveoli into nearby blood vessels. Carbon dioxide moves from the blood into the alveoli and is then exhaled.



Airflow pathway through the respiratory system

Airflow pathway through the respiratory system

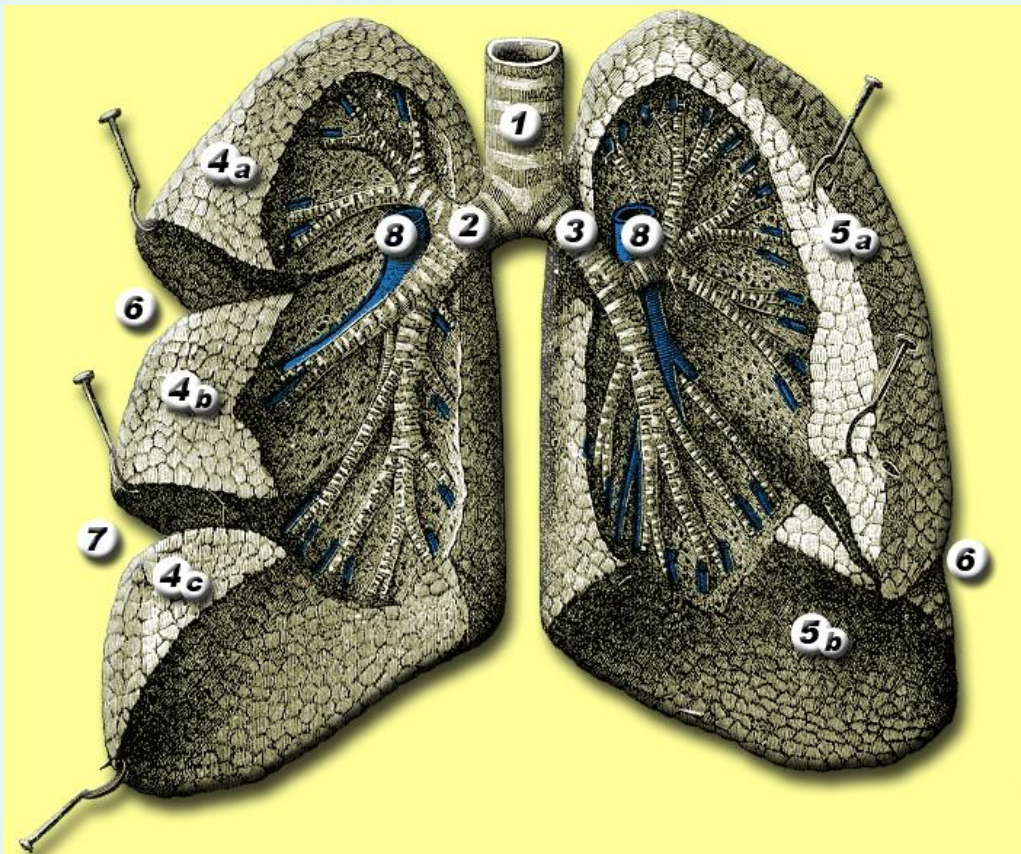
Important Structures

Trachea	A tube that carries air toward the lungs.
Bronchi	The trachea divides into two main bronchi, one leading toward each lung.
Bronchioles	Bronchi divide into smaller branches called bronchioles.
Alveoli	Tiny air sacs where oxygen and carbon dioxide are exchanged between the air and blood.
Right & Left Lungs	Occupy much of the chest cavity and expand and contract during breathing.

EduAR 3D Activity

■ EDUAR 3D TARGET

Insert the Human Lungs AR target image here



EduAR 3D Target — Scan this image to explore the lungs

Explore

- 17. Trachea
- 18. Bronchi
- 19. Right lung
- 20. Left lung
- 21. Bronchioles
- 22. Alveoli

Challenge

Find the alveoli and explain why they are important.

PART 2

Physics

Simple Pendulum

LESSON 4

Understanding a Simple Pendulum

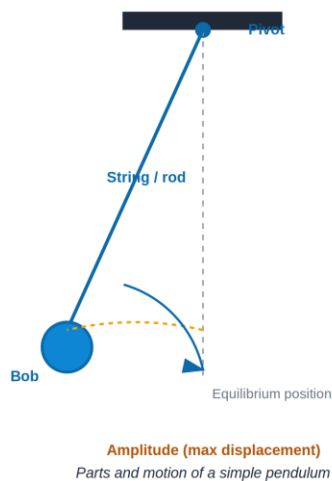
Introduction

A simple pendulum is a physical system consisting of a mass suspended from a fixed point by a string or light rod.

When the mass is displaced from its equilibrium position and released, it swings back and forth. This repeated motion is called oscillation.

The pendulum provides a simple way to explore important physics concepts including:

- Gravity
- Motion
- Oscillation
- Amplitude
- Period
- Frequency



Parts and motion of a simple pendulum

Parts of a Simple Pendulum

Pivot	The fixed point from which the pendulum is suspended.
String or Rod	The part connecting the pivot to the pendulum mass.
Bob	The suspended mass at the end of the pendulum.
Equilibrium Position	The central resting position of the pendulum.
Amplitude	The maximum displacement of the pendulum from its equilibrium position.

How the Pendulum Moves

When the pendulum is pulled to one side, its position changes from equilibrium. When it is released, gravity contributes to the motion that brings it back toward its equilibrium position.

The pendulum continues moving from one side to the other. One complete back-and-forth movement is one oscillation.

Period

The period is the time required for one complete oscillation. The period of a simple pendulum is strongly influenced by its length — a longer pendulum generally has a longer period than a shorter pendulum under comparable conditions.

Frequency

Frequency describes the number of complete oscillations occurring per unit of time. Frequency and period are related:

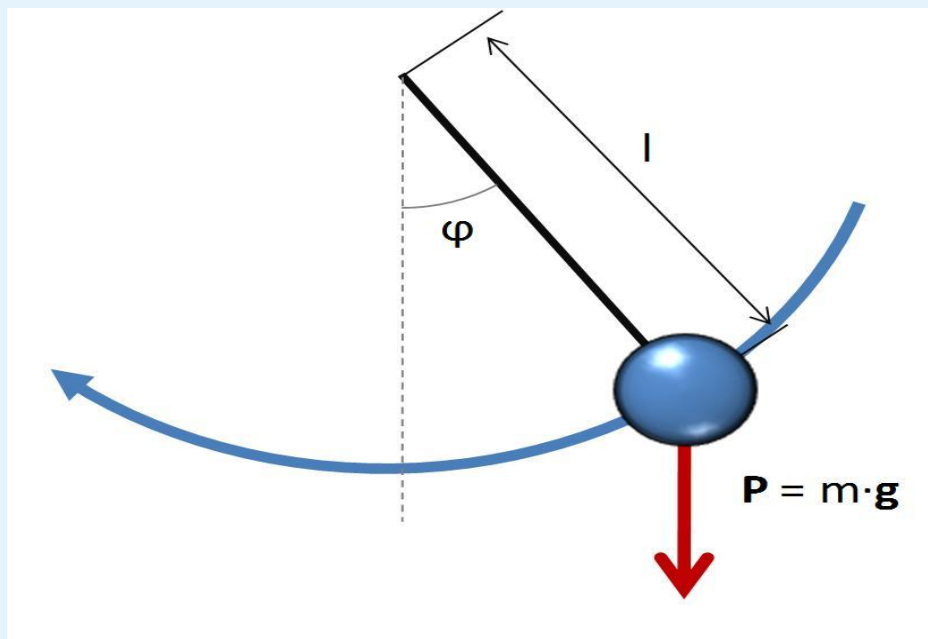
f Formula

Frequency = $1 / \text{Period}$

EduAR 3D Activity

■ EDUAR 3D TARGET

Insert the Simple Pendulum AR target image here



EduAR 3D Target — Scan this image to explore the simple pendulum

Explore

- 23. Pivot
- 24. String or rod
- 25. Bob
- 26. Equilibrium position
- 27. Direction of motion

★ Investigation

Compare a shorter pendulum with a longer pendulum. Ask: which one takes more time to complete one oscillation?

? Think Like a Physicist

- What force causes the pendulum to move back toward its equilibrium position?
- What happens when the pendulum is released from a larger displacement?
- How does pendulum length affect its period?

PART 3

Chemistry

Water Molecule

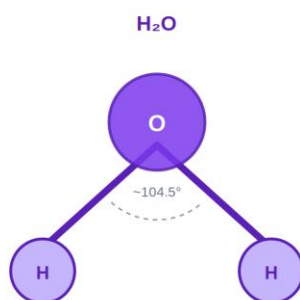
LESSON 5

Understanding H₂O

Introduction

Water is one of the most important substances for life. A water molecule contains three atoms: two hydrogen atoms and one oxygen atom.

Its chemical formula is H₂O. The subscript 2 indicates that two hydrogen atoms are present for every one oxygen atom.



Bent molecular geometry of a water molecule

Bent molecular geometry of a water molecule

Atoms in a Water Molecule

Hydrogen	Two hydrogen atoms are present in each water molecule.
Oxygen	One oxygen atom is present in each water molecule.

Covalent Bonding

The hydrogen and oxygen atoms in a water molecule are connected by covalent bonds. In a covalent bond, atoms share electrons.

The arrangement of the atoms gives water its characteristic bent molecular shape.

Molecular Structure

A simple representation can be shown as $\text{H} - \text{O} - \text{H}$. However, a water molecule is not perfectly straight — its molecular geometry is bent. This three-dimensional arrangement is important for understanding the properties of water.

Why Water Is Important

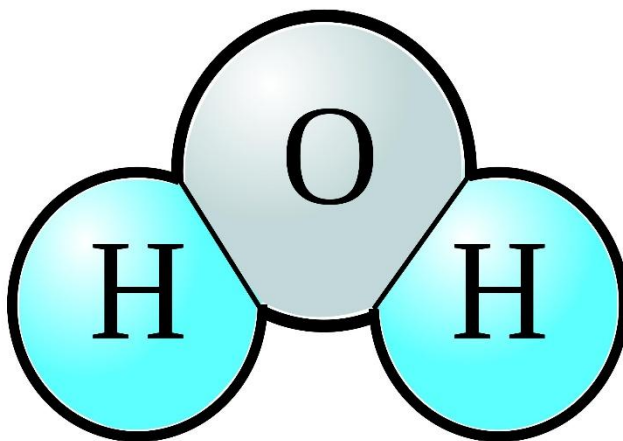
Water plays an important role in:

- Living organisms
- Transportation of substances
- Chemical reactions
- Temperature regulation
- Ecosystems
- Agriculture
- Human activities

EduAR 3D Activity

EDUAR 3D TARGET

Insert the Water Molecule AR target image here



EduAR 3D Target — Scan this image to explore H_2O in 3D

 **Explore**

28. Identify the oxygen atom.
29. Identify both hydrogen atoms.
30. Observe the bonds.
31. Rotate the molecule.
32. View the bent molecular structure.

? Think Like a Chemist

- How many hydrogen atoms are present in a water molecule?
- How many oxygen atoms are present?
- Why is the molecular formula written as H_2O ?
- What type of bond connects the atoms?

PART 4

Mathematics

Three-Dimensional Geometry

LESSON 6

Exploring 3D Shapes

Introduction

Geometry is the study of shapes, sizes, positions and spatial relationships. Two-dimensional shapes have length and width; three-dimensional shapes have length, width, and height.

Three-dimensional geometry helps us understand objects that occupy space. Many everyday objects can be represented using geometric solids.



Cube



Sphere



Cylinder



Cone



Pyramid

Common three-dimensional solids

Common three-dimensional solids

Cube

A three-dimensional solid with six equal square faces.

Properties: Faces: 6 · Edges: 12 · Vertices: 8

Examples: Dice, boxes, building blocks

Sphere

A perfectly round three-dimensional object with one continuous curved surface.

Properties: Flat faces: 0 · Edges: 0 · Vertices: 0

Examples: Ball, globe, spherical containers

Cylinder

Has two parallel circular bases connected by a curved surface.

Properties: Circular bases: 2 · Curved surface: 1

Examples: Cans, pipes, cylindrical containers

Cone

Has one circular base and a curved surface that reaches a single point called the vertex.

Examples: Ice cream cones, traffic cones

Pyramid

Has a polygonal base and triangular faces that meet at a common point called the apex.

Examples: Architectural structures

Real-World Geometry

Geometry is present throughout everyday life: a cube can represent a box, a cylinder a can, a sphere a ball, a cone a traffic cone, and a pyramid certain architectural structures.

Understanding geometry helps students visualize objects in the real world.

Volume

V Volume Formulas

Cube: $V = a^3$ (a = side length)

Cylinder: $V = \pi r^2 h$ (r = radius, h = height)

Sphere: $V = \frac{4}{3} \pi r^3$ (r = radius)

EduAR 3D Activity

EDUAR 3D TARGET

Insert the 3D Geometry AR target image here

(Click to remove this box, then Insert → Picture)

EduAR 3D Target — Scan this image to explore 3D geometry

Explore

33. Rotate the shapes.

34. Identify faces.

- 35. Identify edges.
- 36. Identify vertices.
- 37. Compare different solids.

? Think Like a Mathematician

- Which solid has eight vertices?
- Which solid has no vertices?
- Which solid has two circular bases?
- Which geometric solid has a single apex?

WRAP-UP

Final Review

Four subjects. One learning experience.

BIOLOGY

AR Model(s): Human Heart · Human Brain · Human Lungs

Main learning focus: Human anatomy and biological systems.

PHYSICS

AR Model(s): Simple Pendulum

Main learning focus: Gravity, motion and oscillation.

CHEMISTRY

AR Model(s): Water Molecule — H_2O

Main learning focus: Atoms, bonding and molecular structure.

MATHEMATICS

AR Model(s): 3D Geometry

Main learning focus: Three-dimensional shapes and spatial properties.

AR Learning Checklist

Before Scanning

- ☐ Open the EduAR 3D application.
- ☐ Allow camera access.
- ☐ Make sure the book page is well illuminated.
- ☐ Keep the complete target image visible.

During Exploration

- ☐ Rotate the model.
- ☐ Zoom in.
- ☐ Move the model.
- ☐ Identify important structures.
- ☐ Read the learning information.

After Exploration

- ☐ Answer the quiz.
- ☐ Review incorrect answers.
- ☐ Scan the target again when necessary.

SEE IT. EXPLORE IT. UNDERSTAND IT.

Traditional textbooks explain what something is.

EduAR 3D allows students to see and interact with it.

From the chambers of the heart to the structure of H_2O , from the motion of a pendulum to the geometry of three-dimensional shapes, science becomes a more visual and interactive experience.

Your learning doesn't stop at the page.

READ → SCAN → EXPLORE → LEARN → TEST YOUR KNOWLEDGE